Alireza Jafartash

Senior Back-End Software Engineer

Tehran, Iran | Open to Remote
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Summary

Programming since 2014, from game development to competitive programming and olympiads. Back-end engineer in startups, working across different technologies, programming languages, databases, paradigms, and products. Also builds front ends for full-stack products. M.Sc. candidate in Computer Engineering.

Skills

- **Primary:** Python (Django), Go (Golang), Java 21 (Spring Boot 3), TypeScript (NestJS, Next.js, React, Node.js), PostgreSQL (Query Optimization, Partitioning, Replication), Redis, Apache Kafka, RabbitMQ, REST, Docker, Clean Architecture, SOLID, Concurrency Control, Code Review
- **Familiar:** Kotlin, MongoDB, GraphQL, API Gateways & Rate Limiting, Kubernetes, Terraform, GCP, CI/CD

Work Experience

Software Engineer
LvtGames
Aug 2025–Mar 2026 · Istanbul, Türkiye · Remote · Full-time
- Built high-throughput transaction processing and multi-balance wallet services in Go 1.24 and Spring Boot 3 (Java 21) for a multi-tenant B2B2C entertainment platform.
- Provisioned GCP infrastructure with Terraform, implemented Bitbucket layer-cached CI/CD pipelines that cut deployment times by ~30%, and led cross-functional feature launches.

Tools: Go 1.24, Java 21, Spring Boot 3, PostgreSQL, Redis, GCP, Terraform, Liquibase

Software Engineer
SwapWallet
May 2025–Jul 2025 · Tehran · Full-time
- Resolved critical ledger concurrency race conditions and optimized the crypto exchange matching engine under peak load, reducing p99 execution latency.
- Automated administrative and member operations by designing an event-driven Telegram workflow bot integrated with core exchange services.

Tools: Kotlin, Spring Framework, Apache Kafka, PostgreSQL, Kubernetes

Software Engineer
Ultratrader
Feb 2024–Oct 2024 · Toronto, Canada · Remote · Full-time
- Designed a real-time ingestion pipeline that streams and normalizes financial market data across 30+ external broker APIs at ~1,000 events/sec sustained, with bursts to 300k events/sec.
- Built REST and GraphQL APIs alongside low-latency broker proxy gateways, and optimized analytical PostgreSQL queries to cut reporting latency by more than 90%.

Tools: NestJS, TypeScript, PostgreSQL, Redis, GraphQL, Docker

Software Engineer
Ravindex
May 2022–Dec 2023 · Tehran
- Ingested and processed stock-market tick data through high-throughput analytical pipelines backed by replicated PostgreSQL databases.
- Developed event-driven microservices using Django and RabbitMQ across 20+ automated data-fetching workers, instrumented with Prometheus and Grafana alerting.

Tools: Python, Django, PostgreSQL, RabbitMQ, Docker, Prometheus, Grafana

Research and Development Intern
Plotset
Aug 2020–Jul 2021 · California, US · Remote
- Researched AI-driven data visualization and architected a modular rendering framework integrating multiple charting engines in Node.js and D3.js.

Game Developer
CubeCraft
Sep 2016–Feb 2017 · London, UK · Remote
- Built customizable and maintainable Spigot game mode plugins handling high concurrency.
- Designed state machines with 50+ states and 100+ transitions to manage complex gameplay loops.
- Implemented over 300 event listeners and schedulers to optimize real-time tick performance.
- Debugged and enhanced low-level libraries in the core Java codebase.

Tools: Java, Spigot API, PaperMC, MongoDB

Other Experience

Developer & Technical Product Manager
Manova: Falak
Jul 2026–Present · Sari, Mazandaran, Iran

- Designed and built Falak, a multi-tenant SaaS for restaurant operations covering stock across locations, receipts with partial payments and overdue tracking, and role-based access with location-scoped roles.
- Ran requirement sessions with restaurant owners who were not technical, turning their daily routines into module specifications, acceptance flows, and architecture decisions that set each release's scope.
- Defined and verified the UX rules for staff working on phones and tablets: 390 px and 768 px checked before desktop, 44 px touch targets, full-screen flows on small screens, and Persian and English in RTL at every width.
- Modeled stock as an append-only movement ledger with row-locked atomic batches and negative-balance guards, and enforced tenant isolation with PostgreSQL row-level security (FORCE RLS) on the inventory, receipts, and identity tables.

Tools: Go, PostgreSQL, Next.js, Orbit Auth

Developer & Technical Product Manager
Goldstein
Jul 2026–Present · Mashhad, Khorasan, Iran · Contract

- Co-built a multi-tenant SaaS platform for jewelry retailers to trade physical assets in real time, covering customer verification, a digital wallet ledger, live buy and sell pricing, and physical delivery.
- Ran requirement sessions with store owners who were not technical, converting them into a feature catalog, state machines, and a logged list of open decisions, with a written change process so scope moved only with sign-off.
- Built a concurrency-safe trading engine with pessimistic row locks on inventory and wallet balances, no short selling, and an append-only audit ledger for every balance change.
- Load-tested the engine and traced a cross-store slowdown (one busy store slowed others about 33× at p50) to a shared database connection pool. Sizing the pool and adding PgBouncer brought the slowdown to 1.35× in tests.

Tools: NestJS, Next.js, PostgreSQL, Prisma, Redis, PgBouncer

Head Teaching Assistant — Advanced Programming
University of Tehran
2026–Present · Tehran

- Lead a team of 15+ TAs across problem design, code reviews, grading, and curriculum delivery for 50+ undergraduate students per semester.
- Designed and maintain the containerized Docker CI judge and automated assignment evaluation toolchain used across all terms for Java OOP, SOLID, and concurrency coursework.

Co-Founder & Lead Architect
Hambazi — Boardgame & Startup Community
Aug 2025–Present · Tehran

- Built a Telegram bot for user, event, and game management, backed by the community platform backend (NestJS and Next.js) that supports event scheduling, matchmaking, and member workflows for 1,000+ active members.

Education

M.Sc. in Computer Engineering — Algorithms and Computation
University of Tehran · Sep 2024–Present

Research focus: Large language models, hallucination, and mechanistic interpretability. Thesis: Interpreting Neural Network Thoughts.

B.Sc. in Computer Engineering
Babol Noshirvani University of Technology · Sep 2018–Aug 2024

Thesis: High-Throughput Distributed UUID Generation System — Engineered a collision-free 128-bit (16-byte) distributed ID generator in Go and RabbitMQ sustaining 10,000,000 IDs/sec per node with automated node coordination.

Projects

Promux
Autonomous multiplexer for agentic AI CLIs with zero-downtime account rotation.

Architected a proxy multiplexer that intercepts agentic CLI calls. Automatically detects HTTP 429/500 quota exhaustion, triggers account rotation with TTL cooling, and preserves running agent session context without interrupting long code generation loops.

Go, CLI, AI Systems

Manova / Orbit Platform
Shared foundation for client products: auth, limits, billing, and tenant isolation.

Shared micro-platform for client products, with RBAC, token-bucket rate limits, billing journals, and partitioned PostgreSQL for multi-tenant isolation across 5+ applications.

TypeScript, Go, PostgreSQL, Edge

Dorost

Open-source bilingual document composition package on official Typst registry.

Authored and published an open-source Typst package providing zero-dependency RTL/LTR mixed-direction layout, numeral conversion, and design system components with automated CI test suites.

Typst, Open Source, Design System